Week = 3 27/10/26

3. Iterative Deepening Search (IDS)

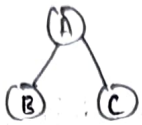
```
ITERATIVE-DEEPING-SEARCH (initial, goal, graph,
                          maximum_depth)
    for depth ← 0 to maximum_depth do
        result ← DEPTH-LIMITED-SEARCH (initial, goal,
                                       graph, depth)
        if result is a solution then
            return result
    return failure

DEPTH-LIMITED-SEARCH (node, goal, graph, limit, path)
    if node = goal then
        return path + [node]
    if limit = 0 then
        return cutoff
    cutoff_occurred ← FALSE
    for each child in graph[node] do
        if child is not in path then
            result ← DEPTH-LIMITED-SEARCH (child,
                      goal, graph, ~~depth~~ limit - 1,
                      path + [node])
            if result is a solution then
                return result
            if result = cutoff then
                cutoff_occurred ← TRUE
    if cutoff_occurred then
        return cutoff
    else
        return failure
```

Test cases :-

1. Goal at Depth 1

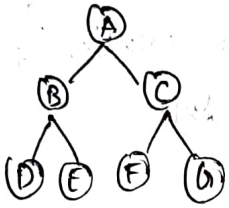
Input:
Initial State : A
Goal State : B
Max Depth : 3



Output:
Goal 'B' found!
Solution Path : A → B
Found at Depth : 1

2. Goal at Depth 2

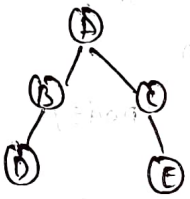
Input:
Initial State : A
Goal State : G
Max Depth : 3



Output:
Goal 'G' Found!
Solution Path : A → C → G
Found at Depth : 2

3. Goal Not Present

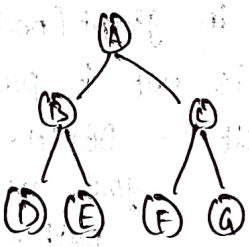
Input:
Initial state : A
Goal state : Z
Max Depth : 2



Output:
Goal 'Z' not found within max depth of 2.

4. Goal at max depth

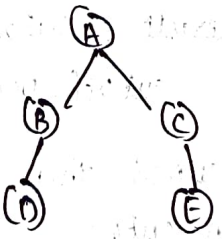
Input:
Initial State : A
Goal state : G
Max Depth : 2



Output:
Goal 'G' found!
Solution Path : A → C → G
Found at Depth : 2

5. Goal Not Present

Input:
Initial State : A
Goal State : Z
Max Depth : 2



Output:
Goal 'Z' not found within max depth of 2.